THE HIDDEN CORE OF ROMANIAN: THE WORDS THAT DEFINE ALL OTHERS

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Abstract

Dictionaries define words by other words, so a dictionary can teach new words only to a reader who already knows part of its vocabulary. Representing a dictionary as a directed graph, in which each word points to the headwords in whose definitions it occurs, makes this part measurable (Vincent-Lamarre et al., 2016). The Kernel is the part from which all other headwords can be defined; a minimal grounding set (MinSet) is a smallest set of words from which every other headword can be learned through definitions alone.

We analyse this graph for the second edition of the DEX (*Dicționarul explicativ al limbii române*), digitised by dexonline: 63,843 headwords and 600,473 definitional links. Because Romanian is richly inflected, many word forms in definitions are ambiguous (*are*: *avea* 'have' or *ara* 'plough'). A large language model assigns each to a headword; all results are repeated without the model, and integer linear programming bounds the MinSet.

The Kernel of the DEX contains 33.2% of the headwords and the MinSet 9.7–9.9%. Processed in the same way, dictionaries of eleven other languages have MinSets of 0.6–3.1%. The difference comes mainly from reciprocal definition, five to twenty times more frequent in the DEX: 8,761 pairs of headwords occur in each other's definitions (*arbore–copac*, both 'tree'). Although derived only from definitions, the Kernel contains 98% of the Romanian equivalents of the Swadesh list, a standard list of about 200 basic meanings used in comparative linguistics, and 94.5% of the 1,419 words that Graur (1954) proposed as the fundamental lexical stock of Romanian, against 33.2% expected by chance. The words most used in definitions, in contrast, are mainly learned French loanwords. Dictionary definitions thus preserve a linguistically meaningful lexical core, and the graph gives lexicographers a map of circularity. Data and code are public.

Keywords: lexicography, dictionary graphs, symbol grounding, minimal grounding set, fundamental lexical stock, Romanian, DEX, large language models